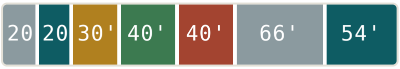
The table of integrals and the rules of integration

Six lessons on the standard integrals, the two linearity rules, and the one substitution that handles every bracket you will meet this year.

LESSON 41–46 | 6 × 40 MIN | ALGEBRA 11, §2.4 | P1 · 10.2–10.3

LESSON CLOCK



The power rule	20 min
The exceptional case	20 min
Linearity	30 min
Rewriting first	40 min
The linear bracket	40 min
Practice and homework	66 min
Consolidation	54 min

LEARNING OBJECTIVES

- Use the power rule for integration, including negative and fractional indices.
- Apply the constant-multiple and sum rules.
- Integrate $(ax + b)^n$.
- Rewrite a quotient or a product into integrable form before integrating.

TEXTBOOK

National	pp. 193–220
Cambridge	Pure Mathematics 1, pp. 203–212
Workbook	P1 Exercise 10B, 10C

01 Explanation

The power rule

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \, , \; n \neq -1$$

Raise the index by one, divide by the new index. It is the power rule for differentiation read backwards, and the check is immediate: differentiating $\frac{x^{n+1}}{n+1}$ gives $\frac{(n+1)x^n}{n+1} = x^n$.

$F(X)$	$\int F(X)\,DX$
x^5	$\frac{x^6}{6} + C$
x	$\frac{x^2}{2} + C$
1	$x + C$
$\sqrt{x} = x^{1/2}$	$\frac{2}{3}x^{3/2} + C$
$\frac{1}{x^2} = x^{-2}$	$-\frac{1}{x} + C$
$\frac{1}{\sqrt{x}} = x^{-1/2}$	$2\sqrt{x} + C$

The one exception

$N = -1$ BREAKS THE RULE

$\int \frac{1}{x}\,dx$ cannot be $\frac{x^0}{0}$ — the division is by zero. The answer is $\ln|x| + C$, which comes from the logarithm, not from the power rule. At this stage, recognise the case and quote the result.

Every other index, positive, negative or fractional, obeys the rule. The exception is a single value, and it is the one every examiner tests.

Linearity

$$\int k\,f(x)\,dx = k\int f(x)\,dx \int [f(x) \pm g(x)]\,dx = \int f\,dx \pm \int g\,dx$$

So a polynomial is integrated term by term, and one constant covers the whole answer:

$$\int (6x^2 - 4x + 5)\,dx = 2x^3 - 2x^2 + 5x + C$$

THERE IS NO PRODUCT OR QUOTIENT RULE FOR INTEGRATION

> $\int f \cdot g\,dx$ is **not** $\int f \cdot \int g$. Test it on $\int x \cdot x\,dx$: the truth is $\frac{x^3}{3}$, but $\frac{x^2}{2} \cdot \frac{x^2}{2} = \frac{x^4}{4}$. Products and quotients must first be **rewritten**.

Rewriting first

Almost every integral at this level is done by turning the integrand into a sum of powers, then applying the rule:

GIVEN	REWRITE AS	INTEGRAL
$\frac{x^3 + 2x}{x}$	$x^2 + 2$	$\frac{x^3}{3} + 2x + C$
$x(x + 3)$	$x^2 + 3x$	$\frac{x^3}{3} + \frac{3x^2}{2} + C$
$(2x - 1)^2$	$4x^2 - 4x + 1$	$\frac{4x^3}{3} - 2x^2 + x + C$
$\frac{1}{x^3}$	x^{-3}	$-\frac{1}{2x^2} + C$
$x\sqrt{x}$	$x^{3/2}$	$\frac{2}{5}x^{5/2} + C$

The linear bracket

The chain rule run backwards gives one more standard result — enough for every bracket whose inside is **linear**:

$$\int (ax + b)^n\,dx = \frac{(ax + b)^{(n+1)}}{a(n + 1)} + C\,,\ n \neq -1$$

The extra a in the denominator undoes the a the chain rule would produce. Check $\int (3x + 1)^4\,dx = \frac{(3x + 1)^5}{15} + C$ by differentiating: $\frac{5(3x+1)^4 \cdot 3}{15} = (3x + 1)^4$ ✓.

> ONLY WHEN THE INSIDE IS LINEAR
>
> $\int (x^2 + 1)^4\,dx$ cannot be done this way — the correction factor would have to be $2x$, which is not a constant. Expand it, or leave it for the next course.

02 Worked examples

EXAMPLE 1 **Find $\int \left(4x^3 - \dfrac{6}{x^2} + \sqrt{x}\right) dx$.**

1. Rewrite: $4x^3 - 6x^{-2} + x^{(1/2)}$.

2. x^4

3. $+ \dfrac{6}{x}$

   $-6 \cdot \dfrac{x^{-1}}{-1}$.

4. $+ \dfrac{2}{3} x^{(3/2)} + C$

ANSWER

$$x^4 + \frac{6}{x} + \frac{2}{3} x\sqrt{x} + C$$

EXAMPLE 2 **Find $\int \dfrac{x^2 - 3x}{x} dx$.**

1. Divide first: $x - 3$.

   No quotient rule exists.

2. $\dfrac{x^2}{2} - 3x + C$

ANSWER

$$\frac{x^2}{2} - 3x + C$$

EXAMPLE 3 **Find $\int (2x + 5)^3\, dx$.**

1. Linear inside, so use the bracket rule.

   $a = 2, n = 3$

2. $$\frac{(2x + 5)^4}{2 \times 4}$$

3. $$= \frac{(2x + 5)^4}{8} + C$$

   Check by differentiating.

ANSWER

$$\frac{(2x + 5)^4}{8} + C$$

03 Interactive model

Differentiate every answer aloud before writing the next question.

Which rule, and what is the answer?

$\int x^4\,dx$:

- $4x^3 + C$
- $\frac{x^5}{5} + C$
- $x^5 + C$
- $\frac{x^3}{3} + C$

$\int \frac{1}{x^2}\,dx$:

- $ln|x| + C$
- $-\frac{1}{x} + C$
- $\frac{1}{x} + C$
- $-2x^{-3} + C$

$\int \frac{1}{x}\,dx$:

- $\frac{x^0}{0}$
- $ln|x| + C$
- 0
- undefined

$\int (3x + 1)^4\, dx$:

$\dfrac{(3x+1)^5}{5} + C$

$\dfrac{(3x+1)^5}{15} + C$

$3(3x+1)^5 + C$

$(3x+1)^5 + C$

$\int x \cdot x\, dx$:

$\dfrac{x^4}{4} + C$

$\dfrac{x^3}{3} + C$

$x^2 + C$

$\dfrac{x^2}{2} + C$

$\int \sqrt{x}\, dx$:

$\dfrac{2}{3}$ x^(3/2) + C

$\dfrac{1}{2\sqrt{x}} + C$

$\dfrac{3}{2}$ x^(1/2) + C

x^(3/2) + C

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Table of integrals	Integrallar jadvali	Таблица интегралов
Power rule for integration	Daraja integrallash qoidasi	Правило интегрирования степени
Constant multiple rule	O‘zgarmas ko‘paytuvchi qoidasi	Правило постоянного множителя
Sum rule	Yig‘indi qoidasi	Правило суммы
Linear substitution	Chiziqli almashtirish	Линейная замена
Integrable form	Integrallanuvchi shakl	Интегрируемый вид
Reciprocal function	Teskari funksiya (1/x)	Обратная величина
Exceptional case	Istisno hol	Особый случай
Term by term	Hadma-had	Почленно

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The power rule for integration fails when:

- $n = 0$
- $n = -1$
- $n = 1$
- never

$\int \frac{1}{x}\, dx$ is:

- $\ln|x| + C$
- $-x^{-2} + C$
- 0
- $x + C$

Is there a product rule for integration?

- yes
- no
- only for polynomials
- only for powers

$\int (ax + b)^n \, dx$ divides by:

- $n + 1$
- a
- $a(n + 1)$
- an

Before integrating $\frac{x^3 + x}{x}$ you should:

- use a quotient rule
- divide through
- square it
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $\int x^3\,dx$

   ANSWER $\frac{x^4}{4} + C$

2. $\int x^6\,dx$

   ANSWER $\frac{x^7}{7} + C$

3. $\int 5\,dx$

   ANSWER $5x + C$

4. $\int 2x\,dx$

   ANSWER $x^2 + C$

5. $\int (x + 1)\,dx$

   ANSWER $\frac{x^2}{2} + x + C$

6. $\int x^{-2}\,dx$

   ANSWER $-\frac{1}{x} + C$

7. $\int \sqrt{x}\,dx$

   ANSWER $\frac{2}{3}$ x^(3/2) + C

Medium · the standard question

7 PROBLEMS

1. $\int (3x^2 - 4x + 1)\,dx$

ANSWER $x^3 - 2x^2 + x + C$

2. $\int \frac{1}{x^3}\,dx$

ANSWER $-\frac{1}{2x^2} + C$

3. $\int \frac{1}{\sqrt{x}}\,dx$

ANSWER $2\sqrt{x} + C$

4. $\int x(x + 2)\,dx$

ANSWER $\frac{x^3}{3} + x^2 + C$

5. $\int \frac{x^3 + 2x}{x}\,dx$

ANSWER $\frac{x^3}{3} + 2x + C$

6. $\int (2x + 5)^3\,dx$

ANSWER $\frac{(2x+5)^4}{8} + C$

7. $\int (1 - x)^5\,dx$

ANSWER $-\frac{(1-x)^6}{6} + C$

Hard · stretch and reason

7 PROBLEMS

1. $\int (2x - 1)^2\, dx$

   ANSWER $\frac{4x^3}{3} - 2x^2 + x + C$

2. $\int x\sqrt{x}\, dx$

   ANSWER $\frac{2}{5} x\text{^}(5/2) + C$

3. $\int (x + \frac{1}{x})^2\, dx$

   ANSWER $\frac{x^3}{3} + 2x - \frac{1}{x} + C$

4. $\int \frac{(x+1)^2}{x^2}\, dx$

   ANSWER $x + 2\ln|x| - \frac{1}{x} + C$

5. $\int \frac{1}{(3x-2)^2}\, dx$

   ANSWER $-\frac{1}{3(3x-2)} + C$

6. $\int \sqrt{4x+1}\, dx$

   ANSWER $\frac{(4x+1)\text{^}(3/2)}{6} + C$

7. Why can $\int (x^2 + 1)^4\, dx$ not use the bracket rule?

   ANSWER The inside is not linear

07 Homework

Homework — 7 tasks

Every answer differentiated back, and $+\ C$ on every line.

1. Find $\int (5x^4 - 3x^2 + 2)\, dx$.

2. Find $\int \left(\sqrt{x} + \frac{1}{x^2}\right) dx$.
3. Find $\int \frac{x^4 - x}{x^2}\, dx$.
4. Find $\int (3x - 4)^5\, dx$.
5. Find $\int (x + 2)(x - 3)\, dx$.
6. Find $\int \frac{1}{\sqrt{2x + 1}}\, dx$.
7. Explain in three sentences why there is no product rule for integration, with an example.